



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 · STANDARD 6.GR.1

Area of Triangles

Flagship

Lesson 5-3-flagship · Teacher Copy

Name: _____

Date: _____

Class: _____

SKY HARBOR MISSION

| The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find the area of a triangle using the formula $A = \frac{1}{2} \times \text{base} \times \text{height}$.

Language Objective: I can explain my steps using the words base, height, area, and perpendicular.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Base

Height

Area

Perpendicular

Composite figure

Formula



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1 The side of a triangle used as the bottom for measuring area is the

.

2 The perpendicular distance from the base to the opposite vertex is the

.

3 The amount of flat space inside a triangle is its .

4 Two lines that meet at a 90° angle are .

5 A shape made of two or more basic shapes combined is a

.

- 6 A math rule written with symbols, like $A = \frac{1}{2} \times b \times h$, is a

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

**Your triangular panel (base 12, height 8) fits inside a 12 by 8 rectangle.
Why do we divide by 2 to find the triangle's area before cutting the glass?**

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Base** — A side of the triangle you use to find the area.

Base — Un lado del triángulo que usas para hallar el área.

- ☐ **Height** — The straight-up distance from the base to the top corner.

Altura — La distancia recta desde la base hasta la esquina de arriba.

- ☐ **Area** — How much space is inside a flat shape.

Área — Cuánto espacio hay dentro de una figura plana.

- ☐ **Perpendicular** — Two lines that meet to make a square corner (90 degrees).

Perpendicular — Dos líneas que se unen formando una esquina recta (90 grados).

- ☐ **Formula** — A math rule written with symbols.

Fórmula — Una regla matemática escrita con símbolos.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

A triangle is half of a ____ with the same base and height.

Un triángulo es la mitad de un ____ con la misma base y altura.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The height is the perpendicular distance from the base to the top corner.

Evidence: Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

Reasoning: This evidence connects the definition of area to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **A triangle is half of a ____ with the same base and height.**

Un triángulo es la mitad de un ____ con la misma base y altura.

- **So the area is $1/2$ times 12 times 8, which equals ____.**

Entonces el área es $1/2$ por 12 por 8, que es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Base
2. **Notes 2:** Height
3. **Notes 3:** Area
4. **Notes 4:** Perpendicular
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **Watch for this mistake:** *A common mistake in Area of Triangles is stopping after multiplying base $\times$ height and forgetting to take half — for example, computing $12 \times 8 = 96$ and writing 96 sq ft instead of 48 sq ft. Another common mistake is using a slanted side as the height instead of the perpendicular (straight-up) distance from the base. Before you submit, ask: Did I take half of base $\times$ height? Is my height measurement perpendicular to the base?*
8. **Try It 1:** A. 42 sq in — *Use the perpendicular height (7 in), not a slanted side. $A = \frac{1}{2} \times 12 \times 7 = \frac{1}{2} \times 84 = 42$ square inches.*
9. **Try It 2:** A. 27 sq ft — *$A = \frac{1}{2} \times 6 \times 9 = \frac{1}{2} \times 54 = 27$ square feet.*